MACHINE LEARNING

ASSIGNMENT – 6

1. In which of the following you can say that the model is overfitting?

C) High R-squared value for train-set and Low R-squared value for test-set.

2. Which among the following is a disadvantage of decision trees?

B) Decision trees are highly prone to overfitting.

3. Which of the following is an ensemble technique?

C) Random Forest
4. Suppose you are building a classification model for detection of a fatal disease where detection of the disease is most important. In this case which of the following metrics you would focus on?

B) Sensitivity

5. The value of AUC (Area under Curve) value for ROC curve of model A is 0.70 and of model B is 0.85. Which of these two models is doing better job in classification?

B) Model B

6. Which of the following are the regularization technique in Linear Regression??

A) Ridge
D) Lasso

7. Which of the following is not an example of boosting technique?

B) Decision Tree

8. Which of the techniques are used for regularization of Decision Trees?

D) All of the above

9. Which of the following statements is true regarding the Adaboost technique?

B) A tree in the ensemble focuses more on the data points on which the previous tree was not performing well

10. Explain how does the adjusted R-squared penalize the presence of unnecessary predictors in the model?

When we add a predictor to a model, the R-squared value will always increase, even if the predictors are not actually improving the accuracy of the model ,also make the model overfitted to the training data.
This is because R-squared only measures how well the model fits the data, not how well it generalizes to new data.

The adjusted R-squared value is calculated as:
Adjusted R-squared = 1 - [(1 - R-squared) * (n - 1) / (n - p - 1)]
Where, n is the sample size and
p is the number of predictors in the model.
The adjusted R-squared value is always less than or equal to the R-squared value, and it decreases as the number of predictors increases.

11. Differentiate between Ridge and Lasso Regression.

Ridge regression

- Ridge regression is a regularization technique, which is used to reduce the complexity of the model.
- It is also called as L2 regularization.
- polynomial regression will fail if there is high collinearity between the independent variables, so to solve such problems, Ridge regression can be used.
- It gives very less importance to the features (but keep that features).

Lasso Regression

- it stands for **Least Absolute Shrinkage and Selection Operator.**
- Some of the features in this technique are completely neglected for model evaluation.
- It is also called as L1 regularization**.**
- Hence, the Lasso regression can help us to reduce the overfitting in the model as well as the feature selection.

12 What is VIF? What is the suitable value of a VIF for a feature to be included in a regression modelling?

VIF stands for Variance Inflation Factor, which is a measure of the degree of multicollinearity in a set of predictor variables in a regression model.
The higher the VIF value, the higher the degree of multicollinearity.

in general, a VIF value of 1 indicates no correlation among the predictor variables,
and values between 1 and 5 are generally considered acceptable.
VIF values greater than 5 indicate that there may be high multicollinearity among the predictor variables,
and values greater than 10 indicate severe multicollinearity that may require remedial action, such as removing one or more of the correlated predictor variables from the model.

Therefore, it is recommended to keep the VIF values as low as possible

13 Why do we need to scale the data before feeding it to the train the model?

Scaling the data before training the model is an important preprocessing step that ensures that the data is on a similar scale and improves the performance and interpretability of the model.

ML algorithms use some form of distance metric to measure the similarity between data points. If the features are on different scales, then the feature with the largest scale will dominate the distance calculations and may lead to incorrect predictions.
Scaling the features ensures that they are on a similar scale, so that distance calculations are meaningful and not dominated by any particular feature.

Many optimization algorithms used in machine learning, such as gradient descent, converge faster when the features are on a similar scale. If the features are on different scales, then the optimization algorithm may take longer to converge or even fail to converge.

Some machine learning algorithms, such as K-nearest neighborS and SVMs, are sensitive to the scale of the features. Scaling the features can improve the performance of these algorithms.

14. What are the different metrics which are used to check the goodness of fit in linear regression?

Metrics that can be used to check the goodness of fit in linear regression.

R-squared (R^2): R-squared is a measure of how well the regression model fits the data, and represents the proportion of variance in the dependent variable that is explained by the independent variables in the model. R-squared values range from 0 to 1, with higher values indicating a better fit.

Mean absolute error (MAE): MAE is a measure of the average absolute difference between the predicted and actual values of the dependent variable. It represents the average amount of error in the predictions, but does not penalize large errors as much as MSE.

Mean squared error (MSE): MSE is a measure of the average squared difference between the predicted and actual values of the dependent variable. It represents the average amount of error in the predictions, and lower values indicate a better fit.

Adjusted R-squared: Adjusted R-squared is a modified version of R-squared that takes into account the number of independent variables in the model. It penalizes models with a large number of independent variables, even if they do not contribute much to the overall fit of the model

15. From the following confusion matrix calculate sensitivity, specificity, precision, recall and accuracy.

Actual/Predicted	True	False
True	1000	50
False	250	1200

Sensitivity/Recall = TP/(TP+FN) = 1000/(1000+50) =1000/1050 = 0.952

Specificity = TN/(TN+FP) = 1200/(1200+250) =1200/1450 = 0.82

Precision = TP/(TP+FP) = 1000/(1000+250) =1000/1250 = 0.80

Accuracy= TP+TN/(TP+TN+FP+FN) = 1000+1200/(1000+1200+50+250) =2200/2500 = 0.88